



DB Schema



Introduction

 Attachment of ERD Rev.1

[attachment:4f343d39-21b7-44bb-8658-94bd9fcff31b:DoctorForms_ERD_R1.pdf](#)

Core Design Principles

- ✓ **Dynamic form structure** (custom templates per user)
- ✓ **Fast retrieval of patient reports**
- ✓ **Multi-template support** (echo, TMT, Cath lab, etc.)
- ✓ **Scalable** for future features: teams, AI autofill, PACS integration
- ✓ Doctors can **switch** between **hospitals** in the app
- ✓ Hospitals see only their **patients** and **reports**
- ✓ **Multi-tenant** SaaS ready
- ✓ **Future role-based** access per hospital (admin, supervisor, doctor, fellow)
- ✓ Flexible billing per hospital if needed later
- ✓ Doctors can create **custom templates** with any fields and structure
- ✓ For each patient report, you store **dynamic data entries** linked to templates
- ✓ Reports generate **PDFs** mapped from report_fields → template_fields
- ✓ Highly scalable for **multi-user clinics or hospitals** in future iterations



Key Tables

1. users

Field	Type	Notes
id	BIGINT, PK	Auto-increment
name	VARCHAR	Cardiologist name
email	VARCHAR, UNIQUE	Login credential
phone	VARCHAR	Optional
password	VARCHAR	Hashed
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

2. patients

Field	Type	Notes
id	BIGINT, PK	Auto-increment
user_id	BIGINT, FK → users.id	Cardiologist owner
hospital_id	BIGINT, FK → hospitals.id	The hospital where patient is registered
name	VARCHAR	Patient name
age	INTEGER	57
dob	DATE	Date of birth (optional)
dos	DATE	date-of-study
gender	ENUM (male, female)	Optional
mrn	VARCHAR	Medical Record Number
height_cm	INTEGER	e.g. 167
weight_kg	INTEGER	e.g. 62
bsa	FLOAT	Body Surface Area (optional auto-calc)
blood_pressure	VARCHAR	"bph/bpl" e.g. "90/73"
referring_physician	VARCHAR	
diagnosis_brief	VARCHAR	brief only
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

3. templates

Ensures template is test-specific (Echo, Treadmill, X-ray) and user-specific.

Field	Type	Notes
id	BIGINT, PK	Auto-increment
user_id	BIGINT, FK → users.id	Template owner
test_id	BIGINT, FK → tests.id	Links template to test type
hospital_id	BIGINT, FK → hospitals.id	If templates are hospital-specific
name	VARCHAR	E.g. Echocardiography Report
description	TEXT	Optional notes
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

4. template_fields

Field	Type	Notes
id	BIGINT, PK	Auto-increment
template_id	BIGINT, FK → templates.id	Belongs to template
section	VARCHAR	Groups fields logically in UI e.g. "Left Ventricle (LV)" / "Header"
label	VARCHAR	Field label, e.g. "LVEF (%)"
type	ENUM (text, number, select, textarea, subtitle, title, image)	Field input type
options	JSON	For select fields: ["Normal", "Abnormal"]
order	INTEGER	Controls field order within the section e.g. 1,2,3 per section
field_group_order	INTEGER	Controls order of sections in UI e.g. 1 (LV), 2 (RV), 3 (Valves)
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

5. reports

Field	Type	Notes
id	BIGINT, PK	Auto-increment

Field	Type	Notes
user_id	BIGINT, FK → users.id	Cardiologist owner
patient_id	BIGINT, FK → patients.id	Linked patient
template_id	BIGINT, FK → templates.id	Based on which template
hospital_id	BIGINT, FK → hospitals.id	The hospital context for this report
test_id	BIGINT, FK → tests.id	For quick lookup
title	VARCHAR	Report title (default from template)
findings	TEXT	Optional structured findings JSON or narrative
conclusion	TEXT	Final impression or conclusion
operator	VARCHAR	Who performed the test
supervisor	VARCHAR	Approving supervisor
device	VARCHAR	e.g. "Vivid E95"
pdf_url	VARCHAR	Path to generated PDF
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

6. report_fields

Field	Type	Notes
id	BIGINT, PK	Auto-increment
report_id	BIGINT, FK → reports.id	Parent report
template_field_id	BIGINT, FK → template_fields.id	Which field
value	TEXT	Entered value
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

7. payments

Field	Type	Notes
Id	BIGINT, PK	Auto-increment
user_id	BIGINT, FK → users.id	Paying cardiologist

Field	Type	Notes
amount	DECIMAL	Subscription fee
status	ENUM (pending, paid, failed)	
payment_provider_id	VARCHAR	Midtrans/Xendit reference
paid_at	TIMESTAMP	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

8. tests

To categorize reports by **test type** rather than only by template. This allows consistent querying, filtering, and future analytics.

Field	Type	Notes
id	BIGINT, PK	Auto-increment
code	VARCHAR	e.g. "TTE"
name	VARCHAR	e.g. "Transthoracic Echocardiography"
type	VARCHAR	e.g. "Invasive, Non-Invasive"
description	TEXT	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

9. measurements

Field	Type	Notes
id	BIGINT, PK	Auto-increment
report_id	BIGINT, FK → reports.id	Parent report
name	VARCHAR	e.g. "IVSd"
value	VARCHAR	e.g. "17.6"
unit	VARCHAR	e.g. "mm, cm/s, m/sec, etc."
category	VARCHAR	"M-Mode", "Doppler"
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

10. hospitals

Field	Type	Notes
id	BIGINT, PK	Auto-increment
name	VARCHAR	Hospital name
address	TEXT	Optional
phone	VARCHAR	Optional
email	VARCHAR	Optional
logo_url	VARCHAR	For branding PDF reports
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

11. hospital_user

Field	Type	Notes
id	BIGINT, PK	Auto-increment
hospital_id	BIGINT, FK → hospitals.id	
user_id	BIGINT, FK → users.id	
role	ENUM(admin, doctor, fellow, supervisor)	Defines access level
department	VARCHAR	Optional (e.g. Cardiology)
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

3 Relationships Overview

- users → templates (1 to many)
- templates → template_fields (1 to many)
- users → patients (1 to many)
- patients → reports (1 to many)
- reports → report_fields (1 to many)
- report_fields → template_fields (many to 1)
- tests → templates → template_fields → report_fields
- patients → reports → report_fields

- reports link directly to tests
- users manage patients, templates, reports
- users ↔ hospitals: Many-to-many via hospital_user.
- hospitals → patients: One-to-many.
- hospitals → reports: One-to-many.
- hospitals → templates: Optional one-to-many (if hospitals define their own templates).

4 Database Relationships

users

- ├— has many templates
- ├— has many patients
- ├— has many reports

templates

- ├— belongs to users
- ├— belongs to tests
- └— has many template_fields

template_fields

- ├— belongs to templates
- └— has many report_fields

tests

- ├— has many templates
- └— has many reports

patients

- ├— belongs to users
- └— has many reports

reports

- ├— belongs to users
- ├— belongs to patients
- ├— belongs to templates
- ├— belongs to tests
- ├— has many report_fields
- └— has many measurements

report_fields

- ├— belongs to reports

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└─ belongs to template_fields
measurements
└─ belongs to reports
hospitals
└─ hospital_user (pivot)
  └─ users
└─ patients
  └─ reports
└─ templates
  └─ template_fields
└─ reports
  └─ report_fields
  └─ measurements

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5 Relationship Details

Table	Relationship	Table	Notes
users	has many	templates, patients, reports	Users manage their own resources
templates	belongs to	users, tests	Templates are user-specific and test-specific
templates	has many	template_fields	Each template has multiple fields
template_fields	belongs to	templates	Field definitions belong to a template
template_fields	has many	report_fields	Each field can have multiple filled values across reports
tests	has many	templates, reports	Test types organize templates and reports
patients	belongs to	users	Each patient is owned by a user
patients	has many	reports	Patients have multiple reports
reports	belongs to	users, patients, templates, tests	Each report links to patient, user, template, and test

Table	Relationship	Table	Notes
reports	has many	report_fields, measurements	Report has both field entries and structured measurements
report_fields	belongs to	reports, template_fields	Field values recorded per report
measurements	belongs to	reports	Measurements are structured per report

6 Simplified Visual Diagram

